Fiscal Year:	FY 2024	**Task Last Updated:**	FY 07/01/2024
PI Name:	Garbino, Alejandro M.D., Ph.D.		
Project Title:	Effect of Carbon Dioxide Exposure on Physical and Cognitive Performance in a Simulated Spaceflight Contingency Scenario (PI: Garbino)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **EVA**:Risk of Mission Impacting Injury and Compromised Performance and Long-Term Health Effects due to EVA Operations		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	01/14/2024	**End Date:**	12/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NOTE: Dr. Alejandro Garbino became Principal Investigator for this project in January, 2024 (Ed., 1/19/24). The previous PI, Dr. Derek Nusbaum, left the laboratory where these experiments were being conducted. Project is a continuation of "Effect of Carbon Dioxide Exposure on Physical and Cognitive Performance in a Simulated Spaceflight Contingency Scenario". Previous PI: Dr. Derek Nusbaum. June 2024 Report: PI notes that "Student Record" section is N/A (Ed., 7/1/24).		
COI Name (Institution):	Abercromby, Andrew Ph.D. (NASA Johnson Space Center) Scully, Robert Ph.D. (KBR/NASA Johnson Space Center) Levine, Benjamin MD (University of Texas Southwestern Medical Center) Babb, Tony Ph.D. (University of Texas Southwestern Medical Center) Pawelczyk, James Ph.D. (Pennsylvania State University) Baughman, Alex M.S. (NASA Johnson Space Center) Bishop, Michael Ph.D. (KBR/NASA Johnson Space Center) Siders, Brett (KBR/NASA Johnson Space Center) Smith, Paul (KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Task Objective/Description (updated by PI in June 2024) Carbon dioxide (CO2) may accumulate inside a space suit during extravehicular activities (EVA). The maximum allowable inspired CO2 during a 1 hour contingency walkback scenario is unknown. The purpose of this study is to characterize the limit of acceptable performance (cognitive and physical) decrements and symptom severity for mission operations when subjected to elevated inspired CO2 levels (up to 30 mmHg). Specific Aim 1 – To quantify operational physical and cognitive performance across a range of inspired CO2 levels during simulated suited contingency EVA operations in a ground-based, shirtsleeve, laboratory study using standardized research techniques. Hypothesis 1.1. – Physical performance (ambulation speed and distance) of simulated EVA tasks will be reduced in a non-linear fashion with increasing partial pressures of inhaled carbon dioxide. Hypothesis 1.2. – Cognitive performance (visual search, spatial memory, paired associate learning) during simulated EVA tasks will be reduced in a non-linear fashion with increasing partial pressures of inhaled carbon dioxide. Specific Aim 2 – To qualify and quantify subjective metrics, including symptoms and self-assessment of performance experienced across a range of inspired CO2 levels during simulated suited contingency EVA operations in a ground-based, shirtsleeve, laboratory study using standardized research techniques. Hypothesis 2.1. – Subjects will experience carbon dioxide-associated symptoms when exposed to elevated partial pressures of inhaled carbon dioxide. Hypothesis 2.2. - Non-linear increases in symptom severity will be observed with increasing partial pressures of inhaled carbon dioxide. Hypothesis 2.3. – Self-assessment of performance will be impaired by increasing partial pressures of inhaled carbon dioxide, despite decrements in objective performance. Methods Study Design - This study will use a repeated measures ground-based study design. A hybrid reality EVA simulation environment will provide a physically and cognitively realistic simulation of a 1 hour contingency walkback, incorporating a passive treadmill and a fully immersive virtual reality simulation of lunar EVA. This study will be performed in a shirtsleeve environment, along with a simulated spacesuit Hard Upper Torso, in a laboratory setting in NASA Johnson Space Center's Building 21 with subjects breathing compressed air mixed with the prescribed fraction of CO2. Subjects will be asked to walk on a treadmill at a self-selected speed within a defined range, bounded by a minimum of 3 km/hr and a maximum speed associated with a maximum allowable metabolic intensity. Cognitive performance will be quantified using validated cognitive tests, as well as measures of functional task performance embedded within the APACHE environment. Physical performance will be measured by evaluating the distance translated by subjects during a one-hour simulated contingency walkback on a treadmill. The severity of specific, CO2 related symptoms (headache, dizziness, nausea, numbness, tingling, shortness of breath, air hunger, fatigue, euphoria) will be quantified via survey prior to the testing session, at 10 minute intervals during the testing session, and immediately following the testing session, as well as every 10 minutes for one hour post exposure to document recovery. Fifteen subjects will be recruited to participate in this study; Each subject will be exposed to 7 different partial pressures of carbon dioxide (0, 5, 10, 15, 20, 25, 30 mmHg) for 1 hour at a time, each on a different day to provide sufficient time for washout of exposure. During the one hour of exposure, the subjects will breathe 21% oxygen compressed air (+CO2) via oronasal facemask. The order in which subjects are exposed to each of the different carbon dioxide partial pressures will utilize balanced randomization and the subjects will be blinded to the pressure of inspired carbon dioxide (PiCO2) levels. The range of CO2 levels to be used have been tolerated in previous studies.
Rationale for HRP Directed Research:	This project is directed because it contains highly constrained research.
Research Impact/Earth Benefits:	Updated by PI in June 2024 The principal benefits of this study are expected to be in the area of operational space medicine, where these data will help inform future planetary EVA suit design and may drive EVA operational constraints. However, as this study deals with the cognitive and physical performance aspects of elevated CO2 exposure, it may have broader implications for clinical patients who are also exposed to elevated CO2, such as patients with chronic obstructive lung disease or patients with obesity hypoventilation syndrome.
Task Progress:	June 2024 Update The breathing apparatus for this study was redesigned so that a similar partial pressure of oxygen (ppO2) would be provided to the subject during CO2 breathing to match that of the partial pressure in the Extravehicular Mobility Unit (EMU) environment during EVAs. The new method for gas delivery to achieve target gas compositions was designed/developed, tested, and implemented. A series of engineering runs were conducted to refine/finalize the test checklist and CO2 blending method (including assessing ease of replicability and consistency across different test days and target CO2 levels). The study is currently recruiting test subjects and collecting data. Data collection is projected to be completed in 2024. New project for FY24 (Ed., 1/19/24). Continuation of "Comparison of the Negative Effects of Carbon Dioxide Exposure Among Ground, Hypobaric, and Spaceflight Environments" under new PI: Alejandro Garbino, M.D., Ph.D. Previous PI: Dr. Derek Nusbaum.
Bibliography Type:	Description: (Last Updated: 04/28/2026)

Abstracts for Journals and Proceedings	Garbino A, Siders B, Keller N, Scully R, Levine B, Babb T, Pawelczyk J, Basner M, Baughman A, Marshall-Goebel K, Abercromby A. "Effect of carbon dioxide exposure on physical and cognitive performance in a simulated spaceflight contingency scenario. Poster." Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024